BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Mid-Term Examination

【Spring Semester, 2021】

Due Date: May 5, 2022 at 14:20 (Problems 4 & 5)

Problem 1: (15%) – In-Class Problem

In this problem we examine a continuous plant-herbivore model. We shall define q as the chemical rate of the plant. Low values of q mean that the plant is toxic; higher values mean that the herbivores derive some nutritious value from it. Consider a situation in which plant quality is enhanced when the vegetation is subjected to a low to moderate level of herbivory, and declines when hervibory is extensive. Assume that herbivores whose density is I are small insects (such as scale bugs) that attach themselves to one plant for long periods of time. Further assume that their growth rate depends on the quality of the vegetation they consume. Typical equations that have been suggested for such a system are

$$\frac{dq}{dt} = K_1 - K_2 q I(I - I_0)$$

$$\frac{dI}{dt} = K_3 I(1 - \frac{K_4 I}{q})$$

1. Explain the equations, and suggest possible meanings for $K_1, K_2, I_0, K_3,$ and K_4.
2. Derive the above differential equations into dimensionless form. Determine the new parameters in terms of original parameters.

Problem 2: (15%) – In-Class Problem

Many biological systems need to be able to respond to changes in the level of some signal (like a hormone) without responding to the actual level. For example, a cell might have no response to a low level of hormone. If the hormone level rapidly increases, the cell responds. But if the hormone level then remains constant at the higher level, the cell again stops responding. The process is sometimes call adaptation.

One mechanism for this process is summarized in the following model. Internal response is a function of the fraction, p, of cell surface receptors that are bound by the hormone. This fraction increases when the hormone level, H, is high. However, the hormone also associate from bound receptors. Assume this happens at a rate A but that this rate is controlled by the cell. One possible set of equation is

$$\frac{dp}{dt} = k_1 H(1-p) - Ap$$

$$\frac{dA}{dt} = e(H-A)$$

Suppose that $k_1 = 0.5$ and the $e = 0.1$, use your computer to simulate the response of the cell.

1. Find the equilibrium state of this model assuming that $H = 1$.
2. Simulate the response when the level of H jumps quickly from $H = 1$ to $H = 10$. Draw graphs of p and A in the phase plane (a plot of state variables p vs. A) as functions of time. Explain what is happening.

Problem 3: (20%) – In-Class Problem

The height (H) of a child is measured at different ages as follows.

age (years)	0	5	8	12	16	18
H (inch)	20.0	36.2	52.0	60.0	69.2	70.0

Estimate the height of the child as an adult of 15 and 30 years of age, using the following two model. You may use MATLAB or python programs to find the parameters of each model. Compare and discuss the two models. Which model do you think is more appropriate to describe the data? Which model is more reasonable for the estimation of the height of an adult of 30 years of age?

Model A: $H = \dfrac{a}{1 + be^{-ct}}$

Model B: $H = \sqrt{a + bt}$

Problem 4: (25%) – Take-Home Problem

May et al. (1979) have proposed the following model to study the interactions between baleen whales and their main food source, krill (a small shrimp-like animal), in the southern ocean:

$$\frac{dx}{dt} = rx\left(1 - \frac{x}{K}\right) - axy$$
$$\frac{dy}{dt} = sy\left(1 - \frac{y}{bx}\right)$$

Here the whale carring capacity is not constant but is a function of the krill population:

$$K_{whales} = bx$$

1. Suggest an interpretation of the terms in these equations after reading the references.
2. Derive the dimensionless form of this model.
3. Run the simulations with the specific parameters: $r = 20$, $a = 5$, $K = 2$, $b = 3$, and $s = 2$, assuming different initial conditions of x and y. Find the equilibrium state.
4. Reproduce the results of Figure 4 and Figure 5 in the paper by May et al. (1979).
5. Try to criticize this model with your own opinions and the related papers you have read.

Note: You will need to consult the following papers to solve this problem:

1. May, R.M., J.R. Beddington, C.W. Clark, S.J. Holt, and R.M. Laws. 1979. Management of multispecies fisheries. Science 205: 267–277.
2. Beddington, J. R. and May, R. M. 1982. The harvesting of interacting species in a natural ecosystem. Scientific American, 247(5): 62–69. http://www.jstor.org/stable/24966726.
3. Beddington, J. R. and Cooke, J. G. Harvesting from a prey-predator complex. Ecological Modeling 14: 155-177.

Problem 5: (25%) – Take-Home Problem

Anderson et al. (1986) proposed a model about AIDS based on the following differential equations.

$$\frac{dX}{dt} = B - \mu X - \lambda c X, \qquad \lambda = \frac{\beta \mathrm{Y}}{\mathrm{N}}$$

$$\frac{dY}{dt} = \lambda c X - (v + \mu) Y,$$

$$\frac{dA}{dt} = p v Y - (d + \mu) A,$$

$$\frac{dZ}{dt} = (1 - p) v Y - \mu Z,$$

$$N(t) = X(t) + Y(t) + Z(t) + A(t)$$

Please refer to the original paper for the notation and meaning of these equations. Answer the following questions after carefully studying the paper.

1. Reproduce the results of the Fig. 10 and Fig. 11 in the original paper of Anderson et al. (1986).
2. By adjusting parameters of the model, one can observe the effect of parameters on the model behavior. Which parameter do you think is the most sensitive parameter in this model? Why?

Note: You will need to consult the following paper to solve this problem:

Anderson, R.M., G.F. Medley, R.M. May, and A.M. Johnson. 1986. A Preliminary study of the transmission dynamics of the human immunodeficiency virus (HIV), the causative agent of AIDS. Math. Med. Biol. 3:229-263.

BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Mid-Term Examination

【Spring Semester, 2020】

Due Date: June 10, 2021 at 17:00 (Problems 4 & 5)

Problem 1: (15%) – In-Class Problem

In the SW deserts of North America, ants, birds, small mammals, and plants interact to create a complex foodweb. The primary interactions are as follows. Ants and small mammals compete for seeds produced by two kinds of plants: small-seeded and large-seeded plants. Within limits, both granivores can consume both sizes of seeds, but, understandably, ants favor small seeds and mammals prefer large seeds. Consumption of seeds reduces the population growth rates of the plants. Birds also consume large seeds, but are more effective at times when the amount of bare ground is high (or, the amount of plants is low). Neither birds nor small mammals eat ants. The two types of plants compete for space.

哺乳動物，

兩個種子捕食者:
螞蟻跟小哺乳類

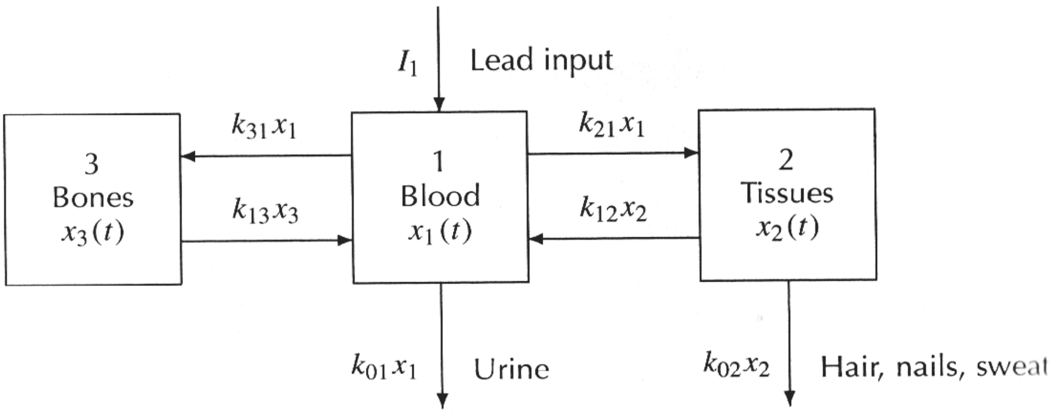
裸地量高時

Draw a Forrester diagram for the population dynamics of these five groups for a model that simulates a period of 20 years at one-month intervals. Assume that both plant types produce seeds in the fall, but that there is a seed pool available to granivores during other months.

Ants
small mammals
small-seeded
large-seeded plants
birds

Problem 2: (15%) – In-Class Problem

Lead is an ingredient in many objects of everyday life: car batteries, water pipes, glassware, ceramics, paint, and gasoline. But lead is toxic, and high levels in the blood and tissues will impair motor and mental capacities. One way to begin to understand this is to build a model of lead flow in the body.



Lead enters the bloodstream via food, air, and water. It accumulates in the blood, in tissues, and especially in the bones. Some lead is excreted by the kidneys and by hair, nails, and sweat. Numbering the body compartments as 1, 2, and 3, the following schematic diagram represents the flow of lead through the compartment. Applying the conservation law to the lead flow through the blood, tissue, and bone compartments, we can model the system with three rate equations:

blood: $\frac{dx_1}{dt} = -(k_{01} + k_{21} + k_{31})x_1 + k_{12}x_2 + k_{13}x_3 + I_1$

tissues: $\frac{dx_2}{dt} = k_{21}x_1 - (k_{02} + k_{12})x_2$

bones: $\frac{dx_3}{dt} = k_{31}x_1 - k_{13}x_3$

Michael Rabinowitz, George Wetherill, and Joel Kopple made a controlled study of the lead intake and excretion of a healthy volunteer living in Southern California. The data from this study were used to estimate the values of the lead intake rate I_1 in micrograms/day and the rate constants k_{ji} in $(\text{days})^{-1}$:

$I_1 = 49.3;\quad k_{01} = 0.0211;\quad k_{21} = 0.0111;\quad k_{31} = 0.0039$

$k_{02} = 0.0162;\quad k_{12} = 0.0124;\quad k_{13} = 0.000035$

Using the above model and parameters, and taking the following two different sets of initial values, perform numerical simulations and plot the lead levels in the bloodstream, tissues, and bones over a period of 800 days and 8000 days, respectively. Discuss your simulation results.

Initial condition 1: $x_1(0) = 0$; $x_2(0) = 0$; $x_3(0) = 0$

Initial condition 2: $x_1(0) = 1800$; $x_2(0) = 1800$; $x_3(0) = 1800$

Problem 3: (20%) – In-Class Problem

The following table shows the data of time evolution of an algal sample taken in the Adriatic Sea (Zangrandi, 1991; Cavallini, 1993):

Time (days)	11	15	18	23	26	31	39	44	54	64	74
Biomass (mm^2)	0.00476	0.0105	0.0207	0.0619	0.337	0.74	1.7	2.45	3.5	4.5	5.09

Estimate the biomass at 150 days, using the following three models. You may use MATLAB software or other programming tools to find the parameters of each model. Compare and discuss the three models. Which model do you think is more appropriate to describe the data? Which model is more reasonable for the estimation of the biomass at 150 days?

Model A: $B = \frac{a}{1+be^{-ct}}$

Model B: $B = a + bx^c$

Model C: $B = \frac{a}{[1+be^{-ct}]^{1/d}}$

Problem 4: (25%) – Take-Home Problem

In modeling the effect of spruce budworm on forest, Ludwig et al. (1978) defined the following set of variables for the condition of the forest:

$S(t)$ = total surface area of trees
$E(t)$ = energy reserve of trees

They consider the following set of equations for these variables in the presence of a constant budworm population B:

$$\frac{dS}{dt} = r_S S(1 - \frac{S}{K_S}\frac{K_E}{E})$$
$$\frac{dE}{dt} = r_E E(1 - \frac{E}{K_E}) - P\frac{B}{S}$$

The factors r, K, and P are to be considered constant.

1. Interpret possible meanings of these equations and parameters.
2. Draw the Forrester diagram of this model.
3. Determine how many steady states exist.
4. Reproduce Fig. 5 in the paper by Ludwig et al. (1978).
5. Interpret what this might imply biologically.

[Note: You have to consult Ludwig et al. (1978) to solve this problem:

Ludwig, D, D. D. Jones, C. S. Holling. 1978. Qualitative Analysis of Insect Outbreak Systems: The Spruce Budworm and Forest. J. Anim. Ecol. 47:315-332.]

Problem 5: (25%) – Take-Home Problem

Anderson et al. (1986) proposed a model about AIDS based on the following differential equations.

$$\frac{dX}{dt} = B - \mu X - \lambda c X, \qquad \lambda = \frac{\beta \mathrm{Y}}{\mathrm{N}}$$

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Please refer to the original paper for the notation and meaning of these equations. Answer the following questions after carefully studying the paper.

1. Reproduce the results of the Fig. 10 and Fig. 11 in the original paper of Anderson et al. (1986).
2. Which parameter do you think is the most sensitive parameter in this model? Why?

[Note: You have to consult the following paper to solve this problem:

Anderson, R.M., G.F. Medley, R.M. May, and A.M. Johnson. 1986. A Preliminary Study of the Transmission Dynamics of the Human Immunodeficiency Virus (HIV), the Causative Agent of AIDS. Math. Med. Biol. 3:229-263.]